Apparent Weight

Answer key

Conventions used throughout

• **Forces:** F_g = gravity (mg, always down), F_N = normal force (the floor or scale pushing up), T = tension (a cable, rope or hanging scale pulling up).

• **Apparent weight** = F_N (what a scale reads). Apparent mass: $F_N = m_{app}g$, so $m_{app} = F_N \div g$. The real mass doesn't change.

• **Signs in the working:** +y is up, with signed numbers: $F_g = -mg$, and a downward acceleration is negative. Directions such as [up] and [down] appear only in final answers.

• **Method (from the lesson):** find the direction of the acceleration FIRST. The net force points the same way.

• $g = 9.81\ m/s^2$. Some worksheet answers use $g = 9.8\ m/s^2$, so a few differ slightly (for example 980 N instead of 981 N). Every step keeps full precision; only final answers are rounded.

ACTIVITY

How does each part of the ride feel?

Elevators do **6** things:

1) Accelerate upwards from the bottom. 2) Move at a constant speed upwards. 3) Slow down as they reach the top. 4) Accelerate Downwards from the top. 5) Move at a constant speed downward. 6) Slow down as they reach the bottom.

Activity: Sort these motions according to how you would feel on the elevator at the time. (Feel heavier than usual / Feel the same weight as usual / Feel lighter than usual)

1) How do you think the normal force compares to the force of gravity in each of these situations?

Step 1 – Find the direction of the acceleration

Acceleration points the way the velocity is changing. Speeding up: a points the same way as the motion. Slowing down: a points opposite the motion. So 1) and 6) have a upward, 3) and 4) have a downward, and 2) and 5) have a = 0.

Step 2 – Compare F_N with F_g

The net force points the same way as a. Only two forces act on you, F_N up and F_g down, so the bigger one is on the side of the acceleration.

Feel heavier than usual	Feel the same weight as usual	Feel lighter than usual
1) Accelerate upwards from the bottom	2) Constant speed upwards	3) Slow down as they reach the top
6) Slow down as they reach the bottom	5) Constant speed downward	4) Accelerate downwards from the top
a up, so $F_N > F_g$	a = 0, so $F_N = F_g$	a down, so $F_N < F_g$

FINAL ANSWER

Heavier: 1) and 6). Same: 2) and 5). Lighter: 3) and 4).

1) Heavier: $F_N > F_g$; same: $F_N = F_g$; lighter: $F_N < F_g$

QUESTION 2

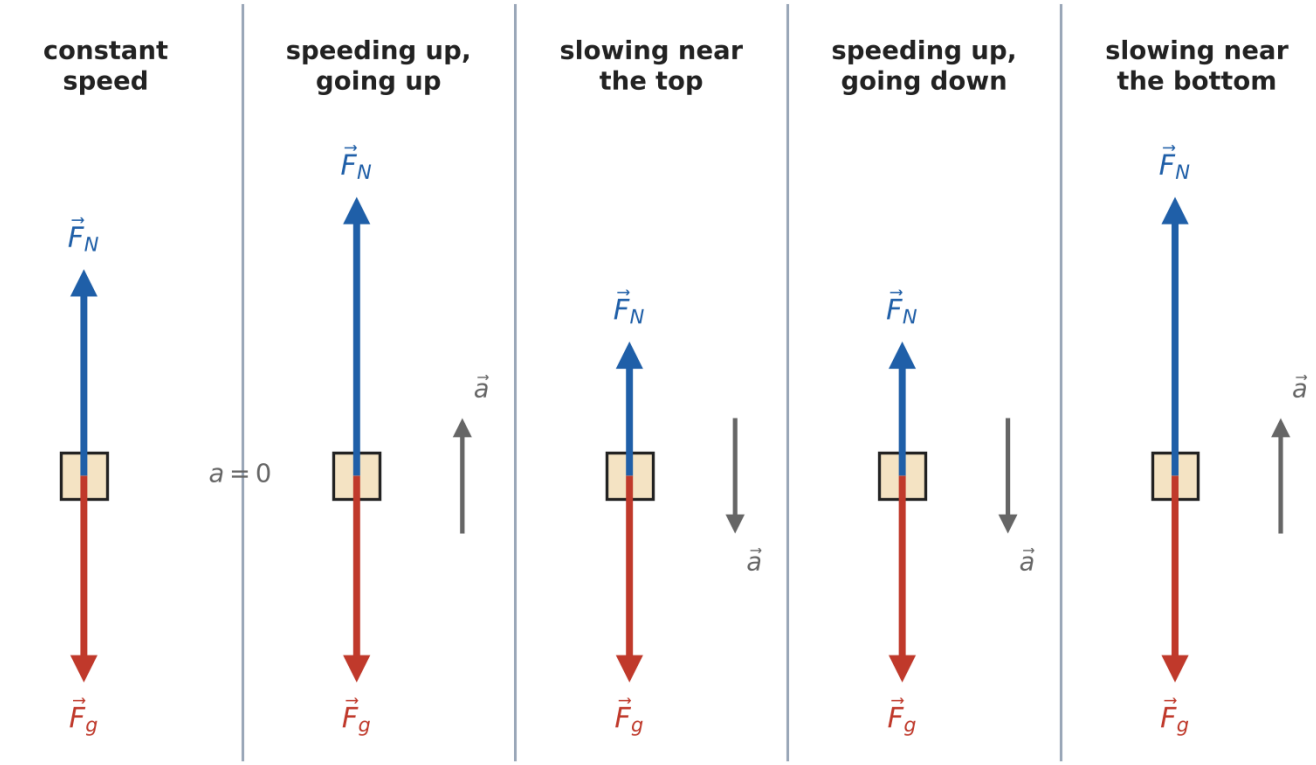
Direction of acceleration and FBDs

2) Fill in the table below. Consider the direction of acceleration FIRST for each situation and make sure your net force has the same direction.

Rows: moving at constant speed upwards or downwards; accelerating upwards from the bottom, ; slowing down as it reaches the top; accelerating downwards from the top; slowing down as it reaches the bottom. Columns: Direction of Acceleration, FBD.

Situation	Direction of acceleration	FBD
moving at constant speed upwards or downwards	none (a = 0)	$F_N = F_g$
accelerating upwards from the bottom,	up	F_N longer than F_g
slowing down as it reaches the top	down (it is moving up but slowing)	F_N shorter than F_g
accelerating downwards from the top	down	F_N shorter than F_g
slowing down as it reaches the bottom	up (it is moving down but slowing)	F_N longer than F_g

FBDs of a person in the elevator (+ up)



F_g is the same in every panel; only F_N changes.

FINAL ANSWER

Constant speed: a = 0. Speeding up going up and slowing near the bottom: a [up], $F_N > F_g$. Slowing near the top and speeding up going down: a [down], $F_N < F_g$.

QUESTIONS 3–5

What the normal force tells us

3) Based on your FBD's, circle the correct statement:

   a. I feel heavier when the force of gravity on me is greater

   b. I feel heavier when the normal force on me is greater

4) Explain **why** the normal force affects how heavy we feel.

5) Circle the best statement:

   a. the normal force is a measure of the force of gravity

   b. the normal force is a measure of the applied force

   c. the normal force is a measure of how much two surfaces are being pressed together

3) Based on your FBD's, circle the correct statement:

In every FBD above, F_g is the same length, but you feel heavier in some rows. The only force that changes is F_N, and you feel heavier exactly when it is bigger. So the answer is b.

4) Explain why the normal force affects how heavy we feel.

Gravity pulls on every part of your body at once, so you can't feel it by itself (in free fall you feel weightless even though gravity still acts). What you feel is the floor pushing up on your feet and your body being squeezed between that push and gravity. That push is F_N: a bigger push feels heavier. A bathroom scale also measures F_N.

5) Circle the best statement:

F_N equals F_g only when a = 0, and it isn't an applied force. It is the push between two surfaces, and it grows the harder they are pressed together. So the answer is c.

FINAL ANSWER

3) b

4) We feel the floor's push (F_N), not gravity itself; a bigger push feels heavier

5) c

CLASS EXAMPLE

Student on a scale in an elevator

1) An elevator is moving upwards, accelerating at 1.2 m/s^2 [down]. A student of mass 65 kg stands on a bathroom scale in this elevator. What is the reading on the scale (in kg)?

To solve the problem, break it down accordingly:

a) draw an fbd

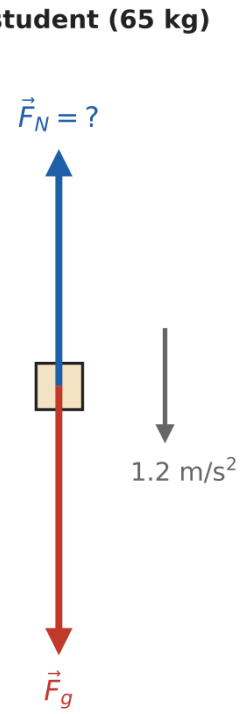
b) determine the normal force on the student [560 kg]

c) determine the apparent mass [57 kg]

a) draw an fbd

The elevator moves up but accelerates down, so it is slowing down near the top. a is down, so F_N is shorter than F_g.

FBD of the student (+ up)



b) determine the normal force on the student [560 kg]

GIVEN

$m = 65$ kg $\qquad a = -1.2$ m/s^2 $\qquad F_N = ?$

Formula: $\Sigma F = ma$

Put in the forces: $F_N + F_g = ma, \quad F_g = -mg$

Add mg to both sides: $F_N = ma + mg$

$F_N = (65)(-1.2) + (65)(9.81) = -78 + 637.65 = 559.65$ N

c) determine the apparent mass [57 kg]

The scale reads in kg by dividing the force on it by g:

Formula: $F_N = m_{app}\, g$

Divide both sides by g: $\quad m_{app} = \dfrac{F_N}{g} = \dfrac{559.65}{9.81} = 57.05$ kg

FINAL ANSWER

(a) FBD above

(b) $\vec{F}_N$ = 5.6 × 10^2 N [up]

(c) The scale reads 57 kg

Worksheet answer: (b) [560 kg]
Why the answers differ: The number is right but the unit is wrong. A normal force is a force, so it is measured in newtons (N), not kilograms: $F_N = 559.65$ N $\approx 5.6 \times 10^2$ N. Kilograms is the unit for part (c), the scale reading $m_{app} = 57$ kg.

PRACTICE 1

Passenger in an elevator

Practice Problems (Do #1-3, 7-9, 11,13)

1. A passenger in an elevator has a mass of 100 kg. Calculate the force, in Newtons, exerted on the passenger by the elevator if the elevator is:

a. at rest (980 N [up])

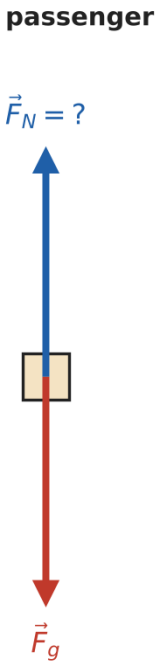
b. moving with an upward acceleration of 15 cm/s^2 (996 N[up])

c. moving with a downward acceleration of 15 cm/s^2 (965 N[up])

d. moving upward with a uniform velocity of 15 cm/s (980 N[up])

e. falling freely (cable breaks) (0 N)

FBD of the passenger (+ up)



The elevator floor pushes on the passenger with F_N. Use $F_N = ma + mg$ (from the Class Example) with 15 cm/s^2 = 0.15 m/s^2:

a. $a = 0: \quad F_N = (100)(0) + (100)(9.81) = 981$ N

b. $a = +0.15: \quad F_N = (100)(0.15) + 981 = 15 + 981 = 996$ N

c. $a = -0.15: \quad F_N = (100)(-0.15) + 981 = -15 + 981 = 966$ N

d. constant velocity, $a = 0: \quad F_N = 981$ N

e. free fall, $a = -9.81: \quad F_N = (100)(-9.81) + (100)(9.81) = 0$ N

In (d) the speed doesn't matter, only the acceleration. In (e) the floor falls with the passenger, so it doesn't push at all.

FINAL ANSWER

(a) 981 N [up]

(b) 996 N [up]

(c) 966 N [up]

(d) 981 N [up]

(e) 0 N

Worksheet answer: (a) 980 N, (c) 965 N, (d) 980 N
Why the answers differ: The worksheet used g = 9.8 m/s^2 in these parts: (100)(9.8) = 980 N and (100)(9.8 − 0.15) = 965 N. This key uses g = 9.81 m/s^2, which gives 981 N and 966 N. The worksheet isn't consistent: its (b) answer, 996 N, uses 9.81 (with 9.8 it would be 995 N). The physics is the same; only the value of g differs.

PRACTICE 2

Tension in an elevator cable

2. An elevator, complete with contents, has a mass of 2000 kg. Determine the value of the tension in the elevator cable when:

a. the elevator is at rest (2.0×10^4 N)

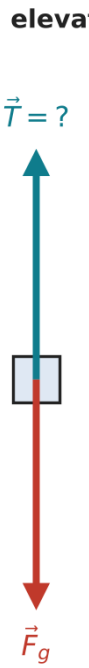
b. the elevator is moving upward at a constant velocity of 2.0 m/s (2.0×10^4 N)

c. the elevator is moving downward at a constant velocity of 2.0 m/s (2.0×10^4 N)

d. the elevator is accelerating upward at 1.0 m/s^2 (2.2×10^4 N)

e. the elevator is accelerating downward at 1.0 m/s^2 (1.8×10^4 N)

FBD of the elevator (+ up)



The cable pulls up with T and gravity pulls down. As before, $T + F_g = ma$, so $T = ma + mg$:

$$\text{a., b., c. } a = 0: \quad T = (2000)(9.81) = 19\,620 \text{ N}$$

$$\text{d. } a = +1.0: \quad T = (2000)(1.0) + 19\,620 = 21\,620 \text{ N}$$

$$\text{e. } a = -1.0: \quad T = (2000)(-1.0) + 19\,620 = 17\,620 \text{ N}$$

Constant velocity, up or down, means a = 0, so (a), (b) and (c) all need the same tension.

FINAL ANSWER

(a), (b), (c) $T = 2.0 \times 10^4$ N

(d) $T = 2.2 \times 10^4$ N

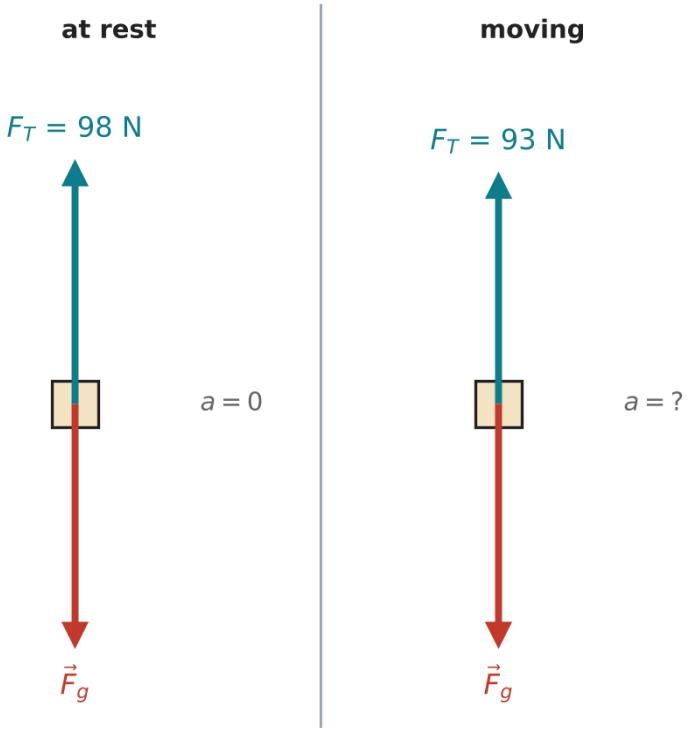
(e) $T = 1.8 \times 10^4$ N

PRACTICE 3

Spring balance on the elevator roof

3. A man measures the acceleration of an elevator by using a spring balance. He fastens the scale to the roof and suspends a mass from it. If the scale reads 98 N when the elevator is at rest and 93 N when the elevator is moving, what is the acceleration of the elevator? (0.50 m/s^2[down])

FBDs of the hanging mass (+ up)



Step 1 – The mass, from the reading at rest

At rest, a = 0, so the scale's pull equals the weight: F_g = 98 N.

Formula: $F_g = mg \quad \Rightarrow \quad m = \frac{F_g}{g} = \frac{98}{9.81} = 9.990 \text{ kg}$

Step 2 – The acceleration while moving

Formula: $\Sigma F = ma$

Put in the forces: $F_T + F_g = ma \quad \Rightarrow \quad 93 + (-98) = (9.990)\,a$

Divide both sides by 9.990 : $a = \dfrac{-5}{9.990} = -0.5005 \text{ m/s}^2$

Negative means down. (The elevator could be moving down and speeding up, or moving up and slowing down.)

FINAL ANSWER

$\vec{a}$ = 0.50 m/s^2 [down]

PRACTICE 4

Block on a table with friction

4. A horizontal force is applied to a 2.0 kg block, moving on a level table. A force that is one-quarter the force of gravity on the block is required to move it at a constant velocity. Calculate the force necessary to accelerate the moving block from rest to a speed of 3.0 m/s in 4.0 s. (6.4 N)

Step 1 – Friction

At constant velocity a = 0, so the push exactly balances friction. Friction is therefore one quarter of the weight:

$$F_f = \frac{1}{4}mg = \frac{1}{4}(2.0)(9.81) = 4.905 \text{ N}$$

Step 2 – The acceleration needed

Formula: $a = \dfrac{v_2 - v_1}{\Delta t} = \dfrac{3.0 - 0}{4.0} = 0.75 \text{ m/s}^2$

Step 3 – The applied force

Take + forward. Friction points backward:

Formula: $\Sigma F = ma$

Put in the forces: $F_{app} - F_f = ma$

Add F_f to both sides:

$$F_{app} = ma + F_f = (2.0)(0.75) + 4.905 = 1.5 + 4.905 = 6.405 \text{ N}$$

FINAL ANSWER

F_{app} = 6.4 N [forward]

PRACTICE 5

Gravity on an unknown planet

5. A space traveler has landed on the surface of an unknown planet similar to earth. He drops a small lead ball from the top of his space ship and finds that it takes 3.0 s to reach the ground 18 m below. If the force of gravity on the astronaut is 710 N on Earth, how much will it be on the planet? (290 N)

Step 1 – The planet's gravitational acceleration

Take + down for the fall. The ball is dropped, so v_1 = 0:

Formula: $\Delta d = v_1 \Delta t + \frac{1}{2} a \Delta t^2$

Put in $v_1 = 0$: $\Delta d = \frac{1}{2} a \Delta t^2$

Multiply by 2 and divide by Δt^2 : $a = \frac{2\Delta d}{\Delta t^2} = \frac{2(18)}{(3.0)^2} = 4.0 \text{ m/s}^2$

Step 2 – The astronaut's mass (from Earth)

$$m = \frac{F_g}{g} = \frac{710 \text{ N}}{9.81 \text{ m/s}^2} = 72.38 \text{ kg}$$

Step 3 – Gravity on the planet

The mass is the same everywhere; only the gravitational acceleration changes:

$$F_{g,\,planet} = m\,a_{planet} = (72.38)(4.0) = 289.5 \text{ N}$$

FINAL ANSWER

$F_g = 2.9 \times 10^2$ N on the planet

PRACTICE 6

Rocket thrust

6. An aerospace scientist has designed a rocket with a mass of 1.0 x 10^3 kg. He wants it to accelerate straight up with an initial acceleration of 21 m/s^2. What thrust must the engine rocket develop? (3.1 x 10^4 N)

Thrust pushes up and gravity pulls down. Take + up:

Formula: $\Sigma F = ma$

Put in the forces: $F_{thrust} + F_g = ma, \quad F_g = -mg$

Add mg to both sides: $F_{thrust} = ma + mg = m(a + g)$

$$F_{thrust} = (1.0 \times 10^3)(21 + 9.81) = 30\,810 \text{ N}$$

FINAL ANSWER

$\vec{F}_{thrust} = 3.1 \times 10^4$ N [up]

PRACTICE 7

Rocket that coasts after its engine shuts off

7. A rocket of mass 1.0×10^3 kg is being fired to a height of 5.0×10^3 m. The rocket engine shuts off when the rocket reaches a height of 1.0×10^3 m, and the rocket coasts up to 5.0×10^3 m.

   a. Draw a free-body diagram to show the forces when, i. the engine is on, and ii. the engine shuts off.

   b. What velocity must the rocket have at the 1.0×10^3 m point to enable it to reach 5.0×10^3 m? (280 m/s)

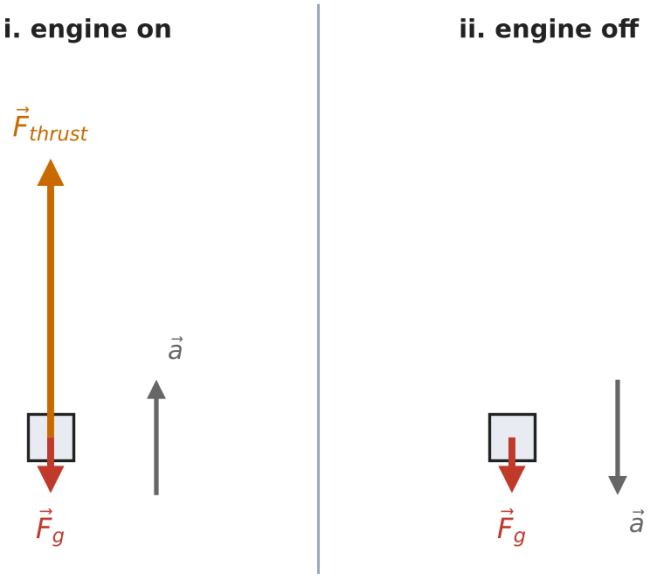
   c. What acceleration did the rocket experience while the engine was on? Off? (39 m/s^2, 9.8 m/s^2)

   d. What force did the rocket engine exert on the rocket? (4.9×10^4 N)

a. Draw a free-body diagram to show the forces when, i. the engine is on, and ii. the engine shuts off.

Engine on: thrust up (bigger than gravity, since the rocket speeds up) and gravity down. Engine off: only gravity (no air resistance), so the rocket slows down as it coasts up.

FBDs of the rocket (+ up)



b. What velocity must the rocket have at the 1.0 x 10^3 m point to enable it to reach 5.0 x 10^3 m?

While coasting from 1.0×10^3 m to 5.0×10^3 m, only gravity acts, so a = -9.81 m/s^2. It just reaches the top, so $v_2 = 0$ there. $\Delta d = 5.0 \times 10^3 - 1.0 \times 10^3 = 4.0 \times 10^3$ m:

Formula: $v_2^2 = v_1^2 + 2a\Delta d$

Put in $v_2 = 0$: $0 = v_1^2 + 2(-9.81)(4000)$

Add 78 480 to both sides: $v_1^2 = 78\,480$

Square root both sides: $v_1 = 280.1$ m/s

c. What acceleration did the rocket experience while the engine was on? Off?

Engine on: from rest at the ground to 280.1 m/s at 1.0×10^3 m:

Formula: $v_2^2 = v_1^2 + 2a\Delta d$

Put in $v_1 = 0$ and divide by $2\Delta d$: $a = \dfrac{v_2^2}{2\Delta d} = \dfrac{78\,480}{2(1000)} = 39.24$ m/s^2

Engine off: only gravity acts, so a = -9.81 m/s^2.

d. What force did the rocket engine exert on the rocket?

As in Practice 6, $F_{thrust} = m(a + g)$:

$$F_{thrust} = (1.0 \times 10^3)(39.24 + 9.81) = 49050 \text{ N}$$

FINAL ANSWER

(a) FBDs above

(b) $\vec{v}$ = 2.8 × 10^2 m/s [up]

(c) On: $\vec{a}$ = 39 m/s^2 [up]; off: $\vec{a}$ = 9.81 m/s^2 [down]

(d) $\vec{F}_{thrust}$ = 4.9 × 10^4 N [up]

Worksheet answer: (c) off: 9.8 m/s^2
Why the answers differ: With the engine off, only gravity acts, so the acceleration is g. The worksheet rounds g to 9.8 m/s^2; this key uses 9.81 m/s^2 throughout. Same answer, different value of g.

PRACTICE 8

Fastest climb on a rope

8. A certain rope can support up to 3 times the mass of a person. Assuming uniform acceleration starting from rest, how long will it take the person to climb to a height of 10.0 m? [1.0 s]

Step 1 – The largest acceleration

The rope can hold the weight of 3 people, so the most it can pull is T = 3mg. The fastest climb uses that pull. Take + up:

Formula: $\Sigma F = ma$

Put in the forces: $T + F_g = ma \quad \Rightarrow \quad 3mg - mg = ma$

Divide both sides by m: $\quad a = 2g = 2(9.81) = 19.62$ m/s^2

Step 2 – The time for 10.0 m

Formula: $\Delta d = v_1 \Delta t + \frac{1}{2} a \Delta t^2$

Put in $v_1 = 0$: $\Delta d = \frac{1}{2} a \Delta t^2$

Multiply by 2, divide by a, square root:

$$\Delta t = \sqrt{\frac{2\Delta d}{a}} = \sqrt{\frac{2(10.0)}{19.62}} = 1.010 \text{ s}$$

Any faster and the rope would break.

FINAL ANSWER

Δt = 1.0 s (the shortest safe time)

PRACTICE 9

Fish on a scale in an elevator

9. A fish hangs from a spring scale in an elevator. As the elevator moves, its weight increases by 10.0%. Determine the acceleration of the elevator. What is the elevator actually doing (moving up / down? Speeding up / slowing down?) [0.981 m/s^2 [up]]

The scale reading (the tension, T) is 10.0% more than mg, so T = 1.100mg. Take + up:

Formula: $\Sigma F = ma$

Put in the forces: $1.100mg - mg = ma$

Simplify: $0.100mg = ma$

Divide both sides by m: $a = 0.100g = (0.100)(9.81) = 0.981\ m/s^2$

a is positive, so it points up. An upward acceleration happens two ways: the elevator is moving up and speeding up, or it is moving down and slowing down. The scale can't tell which.

FINAL ANSWER

$\vec{a}$ = 0.981 m/s^2 [up]

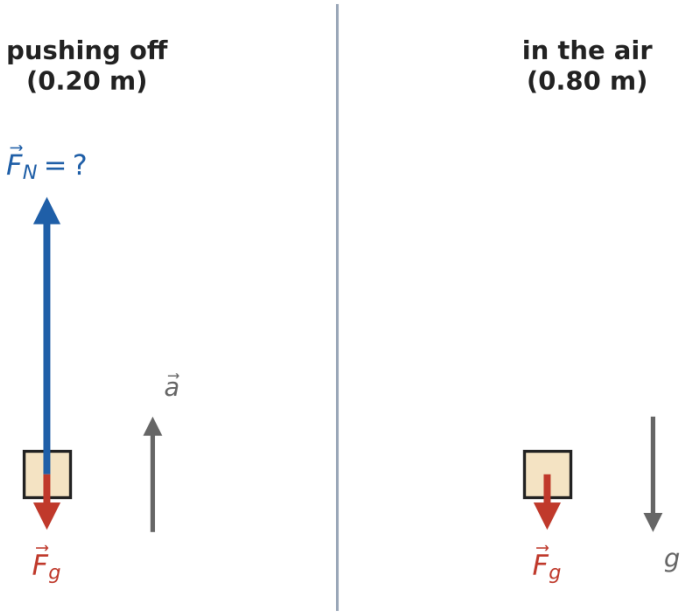
The elevator is either moving up and speeding up, or moving down and slowing down

PRACTICE 10

A high vertical jump

10. An exceptional vertical jump from rest would raise a person 0.80 m off the ground. To do this, what constant force would a 70.0 kg person have to exert against the ground? Assume the person lowers himself by 0.20 m prior to jumping and remains in a standing position while in the air. (3.4 x 10^3 N)

FBDs of the jumper (+ up)



Split the jump into two parts: pushing up off the ground over 0.20 m, then flying 0.80 m up with only gravity acting.

Step 1 – Take-off speed (from the flight)

In the air a = -9.81 m/s^2, and at the top $v_2 = 0$:

Formula: $v_2^2 = v_1^2 + 2a\Delta d$

Put in $v_2 = 0$: $0 = v_1^2 + 2(-9.81)(0.80)$

Add 15.696 to both sides: $v_1^2 = 15.696$ m^2/s^2

Step 2 – Acceleration while pushing

During the push the person goes from rest to that speed over 0.20 m:

Formula: $v_2^2 = v_1^2 + 2a\Delta d$

Put in $v_1 = 0$ and divide by $2\Delta d$: $a = \dfrac{v_2^2}{2\Delta d} = \dfrac{15.696}{2(0.20)} = 39.24\ \text{m/s}^2$

Step 3 – The push from the ground

The ground pushes up with F_N, and by Newton's third law the person pushes down on the ground with the same size:

Formula: $\Sigma F = ma \quad \Rightarrow \quad F_N + F_g = ma$

Add mg to both sides:

$$F_N = m(a + g) = (70.0)(39.24 + 9.81) = 3433.5\ \text{N}$$

FINAL ANSWER

F = 3.4 × 10^3 N (the person pushes down on the ground with this force)

PRACTICE 11

Spider on a strand

11. A 0.10 g spider is descending on a strand that supports it with a force of 5.6 x 10^{-4} N. What is the acceleration of the spider? Ignore the factor of air resistance. (4.2 m/s^2 [down])

Change the mass to kilograms: 0.10 g = 1.0×10^{-4} kg. Take + up:

$$F_g = -mg = -(1.0 \times 10^{-4})(9.81) = -9.81 \times 10^{-4} \text{ N}$$

Formula: $\Sigma F = ma \quad \Rightarrow \quad T + F_g = ma$

Divide both sides by m: $\quad a = \dfrac{5.6 \times 10^{-4} + (-9.81 \times 10^{-4})}{1.0 \times 10^{-4}}$

$$a = \frac{-4.21 \times 10^{-4}}{1.0 \times 10^{-4}} = -4.21 \text{ m/s}^2$$

FINAL ANSWER

$\vec{a}$ = 4.2 m/s^2 [down]

PRACTICE 12

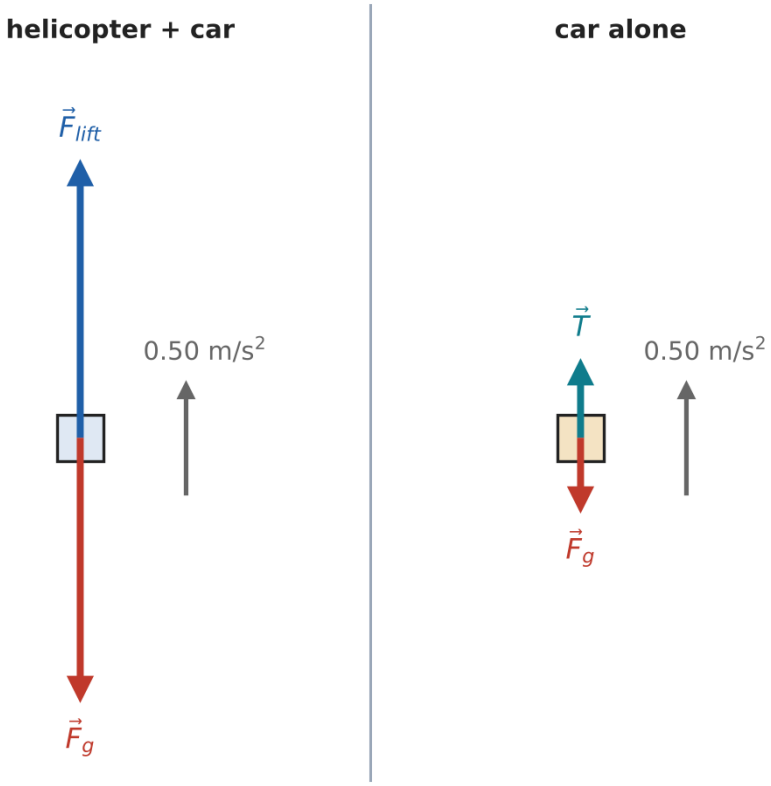
Helicopter lifting a car

12. A 5000 kg helicopter accelerates upward at 0.50 m/s^2 while lifting a 2000 kg car.

a. What is the lift force exerted by the air on the rotors? (7.2 x 10^4 N [up])

b. What is the tension in the cable that connects the car to the helicopter? (2.1 x 10^4 N [up])

FBDs (+ up)



a. What is the lift force exerted by the air on the rotors?

Take the helicopter and car as one 7000 kg system. The cable is inside it and cancels:

Formula: $F_{lift} + F_g = ma \quad \Rightarrow \quad F_{lift} = m(a+g)$

$F_{lift} = (5000 + 2000)(0.50 + 9.81) = (7000)(10.31) = 72\,170$ N

b. What is the tension in the cable that connects the car to the helicopter?

The cable is the only upward force on the car:

$$T = m_{car}(a + g) = (2000)(10.31) = 20\,620 \text{ N}$$

FINAL ANSWER

(a) $\vec{F}_{lift}$ = 7.2 × 10^4 N [up]

(b) T = 2.1 × 10^4 N

PRACTICE 13

Jane climbs the vine

13. Jane wishes to quickly scale a slender vine to visit Tarzan in his treetop hut. The vine is known to safely support the combined weight of Tarzan, Jane and Cheeta. Tarzan has twice the mass of Jane, who has twice the mass of Cheeta. If the vine is 60 m long, what minimum time should Jane allow for the climb? (2.2 s)

Step 1 – The masses

Call Cheeta's mass m. Jane is 2m and Tarzan is 4m, so together they are m + 2m + 4m = 7m. The vine can pull with at most T = 7mg.

Step 2 – Jane's largest acceleration

Jane (mass 2m) climbs with the vine pulling as hard as it safely can. Take + up:

Formula: $\Sigma F = ma \quad \Rightarrow \quad T + F_g = (2m)a$

Put in $T = 7mg,\ F_g = -2mg$: $\quad 7mg - 2mg = 2ma$

Divide both sides by $2m$: $\quad a = \dfrac{5g}{2} = \dfrac{5(9.81)}{2} = 24.525 \text{ m/s}^2$

Step 3 – The time for 60 m

Formula: $\Delta d = v_1 \Delta t + \frac{1}{2} a \Delta t^2 \quad (v_1 = 0)$

$$\Delta t = \sqrt{\frac{2\Delta d}{a}} = \sqrt{\frac{2(60)}{24.525}} = 2.212 \text{ s}$$

FINAL ANSWER

Δt = 2.2 s (minimum)

PRACTICE 14

Scale reading while accelerating down

14. An 80 kg man is standing in an elevator on a set of spring scales calibrated in Newtons. Suppose the elevator accelerates downward at 3.0 m/s^2. What reading will the scales have? (544 N)

The scale reads F_N. Take + up, so a = −3.0 m/s^2:

$$F_N = ma + mg = (80)(-3.0) + (80)(9.81) = -240 + 784.8 = 544.8 \text{ N}$$

FINAL ANSWER

The scales read 545 N

Worksheet answer: 544 N
Why the answers differ: The worksheet used g = 9.8 m/s^2: (80)(9.8 − 3.0) = 544 N. With g = 9.81 m/s^2 the reading is (80)(9.81 − 3.0) = 544.8 N, which rounds to 545 N. (To the 2 significant figures the data really allow, both are 5.4 × 10^2 N.)

PRACTICE 15

Empty elevator on a cable

15. An empty elevator of mass 2.7 x 10^3 kg is pulled upward by a cable at an acceleration of 1.2 m/s^2.

a. What is the tension in the cable? (3.0 x 10^4 N)

b. What would the tension be if the elevator were accelerating downward at 1.2 m/s^2. (2.3 x 10^4 N)

c. Does the direction of motion of the elevator matter in a. and b.?

As in Practice 2, T = ma + mg with + up:

$$\text{a. } a = +1.2: \quad T = (2700)(1.2 + 9.81) = (2700)(11.01) = 29\,727 \text{ N}$$

$$\text{b. } a = -1.2: \quad T = (2700)(-1.2 + 9.81) = (2700)(8.61) = 23\,247 \text{ N}$$

c. No. Only the direction of the acceleration appears in the equation. An elevator moving up while slowing down has the same downward acceleration, and so the same tension, as one moving down while speeding up.

FINAL ANSWER

(a) T = 3.0 × 10^4 N

(b) T = 2.3 × 10^4 N

(c) No: only the direction of the acceleration matters, not the direction of motion

PRACTICE 16

Fish on a scale from the roof

16. A fish hangs from a spring scale supported from the roof of an elevator.

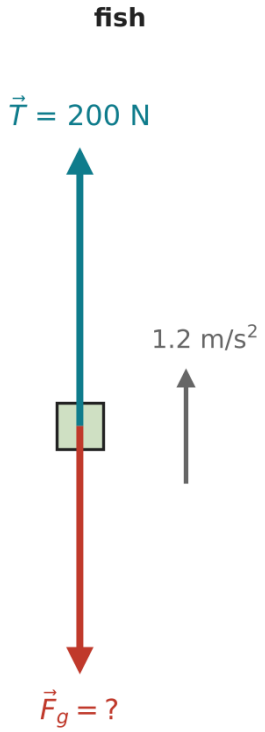
a. If the elevator has an upward acceleration of 1.2 m/s^2 and the scale reads 200 N, what is the true force of gravity on the fish? (1.8 x 10^2 N)

b. Under what circumstances will the scale read 150 N? (1.5 m/s^2 [down])

c. What will the scale read if the elevator cable breaks? (0 N)

a. If the elevator has an upward acceleration of 1.2 m/s^2 and the scale reads 200 N, what is the true force of gravity on the fish?

FBD of the fish (+ up)



The scale pulls up with T = 200 N. Find the mass first, then its weight:

Formula: $T + F_g = ma \quad \Rightarrow \quad T = m(a+g)$

Divide both sides by $(a+g)$: $\quad m = \dfrac{T}{a+g} = \dfrac{200}{1.2+9.81} = 18.165 \text{ kg}$

$F_g = mg = (18.165)(9.81) = 178.2 \text{ N}$

b. Under what circumstances will the scale read 150 N?

Use the same mass with T = 150 N:

$$\text{Formula: } T + F_g = ma \quad \Rightarrow \quad a = \frac{T - mg}{m}$$

$$a = \frac{150 - 178.2}{18.165} = -1.553 \text{ m/s}^2$$

The acceleration is 1.55 m/s^2 down (moving down and speeding up, or moving up and slowing down).

c. What will the scale read if the elevator cable breaks?

Everything falls freely with a = −9.81 m/s^2:

$$T = m(a + g) = m(-9.81 + 9.81) = 0 \text{ N}$$

FINAL ANSWER

(a) F_g = 1.8 × 10^2 N

(b) When $\vec{a}$ = 1.6 m/s^2 [down] (moving down and speeding up, or moving up and slowing down)

(c) 0 N

Worksheet answer: (b) 1.5 m/s^2 [down]
Why the answers differ: The unrounded value is 1.55 m/s^2 (with g = 9.81 it is 1.553; with g = 9.8 it is exactly 1.550). To 2 significant figures, 1.55 rounds **up** to 1.6 m/s^2. The worksheet's 1.5 rounds it down, which is a rounding mistake.